

Question Bank for Basic Electronics (BBEE103/203)

Module – 2

Bipolar Junction Transistors

1. Explain various currents and voltages flowing through the BJT transistor. *(MQP '22 – 7M)*
2. Define α and β . Determine the relationship between α and β . *(MQP '22 – 6M)*
3. A transistor has $\beta = 150$ and I_E is 12 mA. Calculate the approximate collector current (I_C) and base current (I_B). *(Feb '23 – 4M)*
4. Calculate α_{dc} and β_{dc} for the transistor if I_C is measured as 1 mA and I_B is 25 μ A.
5. Describe with neat circuit diagram of BJT amplification for both voltage and current. *(Feb '23 – 8M)*
6. Explain input and output characteristics of the common emitter configuration. *(MQP '22 – 6M)*
7. Briefly explain common collector input characteristics. *(Feb '23 – 4M)*
8. Draw the DC load line for transistor and identify Q point. *(MQP '22 – 8M)*

Field Effect Transistors

1. Explain the transfer and drain characteristics of a JFET.
2. Explain the transfer and drain characteristics of enhancement type MOSFET.
3. Explain depletion type MOSFET along with the transfer and drain characteristics. *(Feb '23 – 8M)*

Module – 3

Operational Amplifiers

1. What is an operational amplifier? Mention its applications.
2. Explain block diagram representation of typical op-amp. *(Feb '23 – 8M, MQP '22 – 6M)*
3. Describe block diagram representation of an op-amp. Also describe its operational behavior with an equivalent circuit. *(MQP '22 – 8M)*
4. Explain the following op-amp parameters
(i) CMRR (ii) Slew rate (iii) Input offset voltage (iv) Input bias current *(Feb '23 – 8M)*
5. A certain op-amp has an open loop voltage gain of 1,00,000 and a common mode gain of 0.2. Determine the CMRR and express it in decibels.
6. Briefly discuss the ideal characteristics of the op-amp. *(MQP '22 – 7M)*

Op-Amp Applications

1. Explain op-amp as an inverting and non-inverting amplifier. *(Feb '23 – 8M)*

2. An inverting amplifier using op-amp has a feedback resistor of $10\text{ k}\Omega$ and one input resistor of $1\text{ k}\Omega$. Calculate the gain of the op-amp and the output voltage if it is supplied with an input of 0.5 V . **(MQP '22 – 4M)**
3. An inverting amplifier has $R_1 = 20\text{ k}\Omega$, $R_f = 100\text{ k}\Omega$. Find the output voltage, input resistance and input current for an input voltage of 1 V .
4. A non-inverting amplifier has closed loop gain of 25. If input voltage $V_i = 10\text{ mV}$, $R_f = 10\text{ k}\Omega$, determine the value of R_1 and output voltage V_o .
5. Explain working of a differential amplifier. **(MQP '22 – 8M)**
6. Briefly explain op-amp as a voltage follower. **(Feb '23 – 4M)**
7. Derive an expression for the 3-input summing circuit. **(MQP '22 – 6M)**
8. Calculate the output voltage of a three-input inverting summing amplifier, given $R_1 = 200\text{ k}\Omega$, $R_2 = 250\text{ k}\Omega$, $R_3 = 500\text{ k}\Omega$, $R_f = 1\text{ M}\Omega$, $V_1 = -2\text{ V}$, $V_2 = -1\text{ V}$ and $V_3 = +3\text{ V}$.
9. Design an adder circuit using an op-amp to give the output $V_o = -(3V_1 + 4V_2 + 5V_3)$. Assume $R_f = 120\text{ k}\Omega$. **(Feb '23 – 4M)**
10. Design the circuit diagram for the output voltage $V_o = -5(0.1V_1 + 0.2V_2 + 10V_3)$. Also draw the neat circuit diagram. **(MQP '22 – 7M)**
11. Develop a summer circuit using op-amp to get the following output voltage $V_o = -(2V_1 + 2V_2)$. **(MQP '22 – 4M)**
12. Design an adder circuit using op-amp to obtain an output voltage, $V_o = -[2V_1 + 3V_2 + 5V_3]$. Assume $R_f = 10\text{ k}\Omega$. With a neat circuit diagram, explain the subtractor using an op-amp.
13. Derive an expression for integrator and differentiator. **(MQP '22 – 6M)**